

Deep Learning

12. Recurrent Neural Networks (contd.)

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Jan-May 2026

So far...

① Perceptron, MLP, Gradient Descent (Backpropagation)

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- ② CNNs

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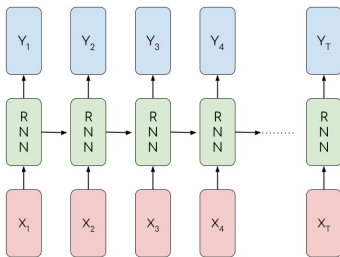
- ① Perceptron, MLP, Gradient Descent (Backpropagation)
- ② CNNs
- ③ 'Feedforward Neural networks'

So far...

- ① Perceptron, MLP, Gradient Descent (Backpropagation)
- ② CNNs
- ③ 'Feedforward Neural networks'
- ④ Sequence learning problems, Simple RNNs

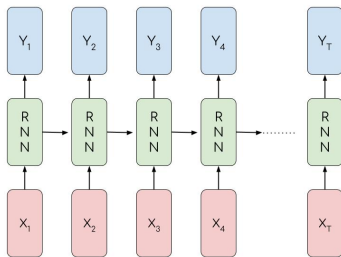
How to train such networks?

① Most generic setting



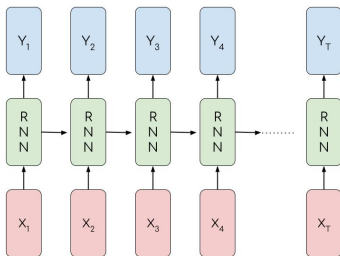
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- ① Most generic setting
- ② I/p: sequence $\{\mathbf{X}_i\}_{i=1}^T$



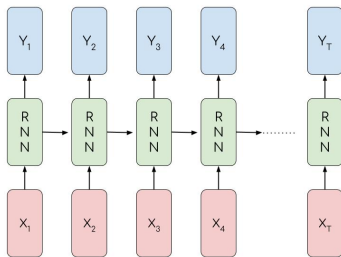
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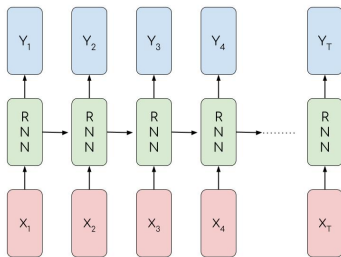
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- ② I/p: sequence $\{\mathbf{X}_i\}_{i=1}^T$
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- ④ Target: sequence $\{\mathbf{D}_i\}_{i=1}^T$



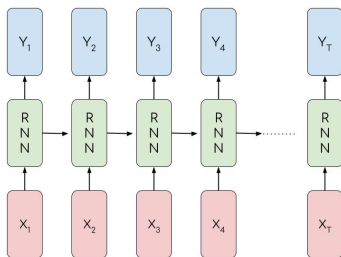
How to train such networks?

- ① Most generic setting
- ② I/p: sequence $\{\mathbf{X}_i\}_{i=1}^T$
- ③ O/p: sequence $\{\mathbf{Y}_i\}_{i=1}^T$
- ④ Target: sequence $\{\mathbf{D}_i\}_{i=1}^T$
- ⑤ Network parameters need to be learned via minimizing an error b/w $\{\mathbf{Y}_i\}_{i=1}^T$ and $\{\mathbf{D}_i\}_{i=1}^T$



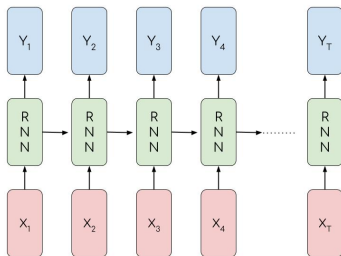
How to train such networks?

- 1 Note that the computation shares the same parameters across time



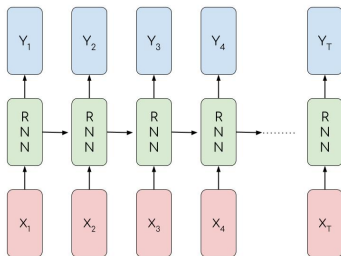
How to train such networks?

- ① Note that the computation shares the same parameters across time
- ② i.e., all columns use the same parameters (green box)



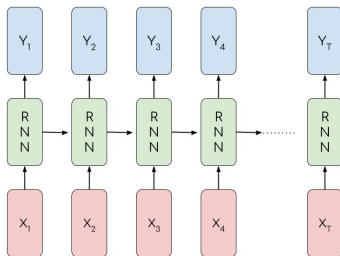
How to train such networks?

- ① Note that the computation shares the same parameters across time
- ② i.e., all columns use the same parameters (green box)
- ③ We use Gradient Descent for these shared parameters (i.e., pooling of the gradients)



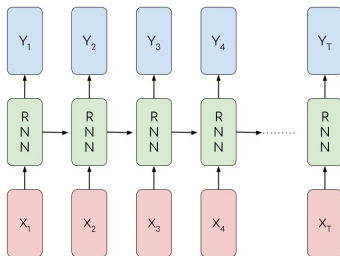
The forward pass

- 1 For each training input:



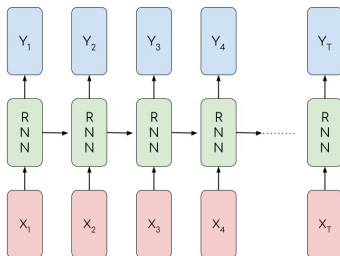
The forward pass

- ① For each training input:
- ② Pass the entire data sequence through the network and generate the outputs



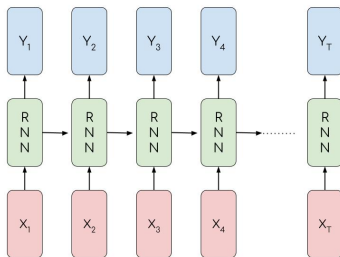
The Gradient computation

① For each training input:



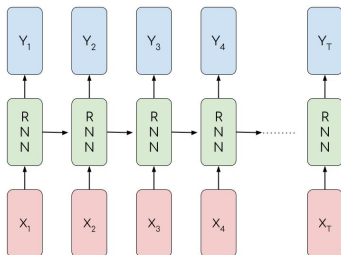
The Gradient computation

- 1 For each training input:
- 2 Compute gradients via backpropagation



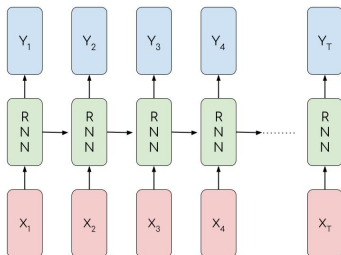
The Error

- ① Error is computed b/w two sequences $\{\mathbf{Y}_i\}_{i=1}^T$ (predicted) and $\{\mathbf{D}_i\}_{i=1}^T$ (target)



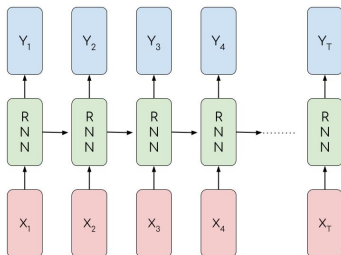
The Error

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- ② Scalar function of series of vectors! (Need not be just the sum of individual errors at individual time steps)



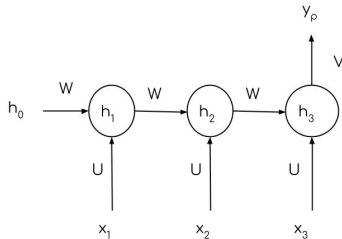
The Error

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- ② Scalar function of series of vectors! (Need not be just the sum of individual errors at individual time steps)
- ③ Note that those sequences can be of different length!



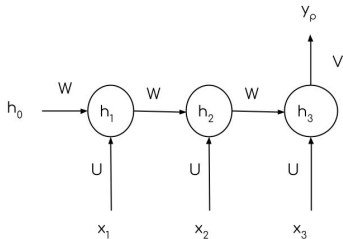
Backpropagation Through Time (BPTT)

- 1 Consider a many-to-one variant RNN (e.g. sentiment analysis)



Backpropagation Through Time (BPTT)

- 1 Consider a many-to-one variant RNN (e.g. sentiment analysis)
- 2 Let's separate the parameters into U , V , and W

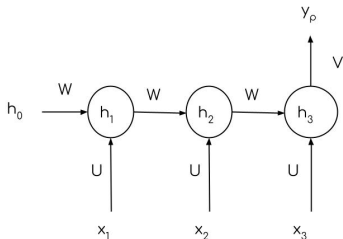


Backpropagation Through Time (BPTT)



భారతీయ టెక్నోలాజికల్ హైదరాబాద్
భారతీయ ప్రాచీనగణిత సంస్థ హైదరాబాద్
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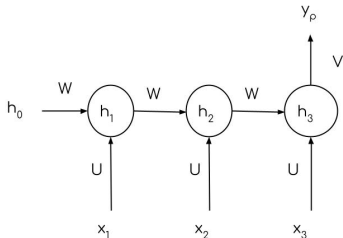
- 1 Let's now perform SGD
(assume loss L is
formulated on y_p)



Backpropagation Through Time (BPTT)



- 1 Let's now perform SGD
(assume loss L is
formulated on y_p)
- 2 $\rightarrow$ we need to compute
 $\frac{\partial L}{\partial V}$, $\frac{\partial L}{\partial W}$, and $\frac{\partial L}{\partial U}$

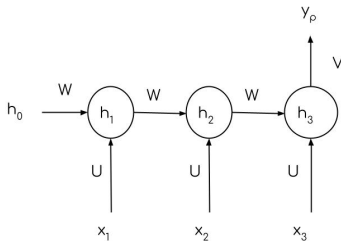


Backpropagation Through Time (BPTT)



భారతీయ టెక్నోలజీ విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

$$\textcircled{1} \quad \frac{\partial L}{\partial V} = \frac{\partial L}{\partial y_p} \frac{\partial y_p}{\partial V} =$$
$$\frac{\partial L}{\partial y_p} \cdot \frac{\partial y_p}{\partial z_3} \cdot \frac{\partial z_3}{\partial V}$$



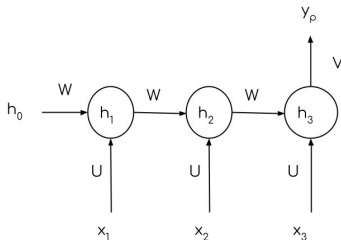
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$$\textcircled{2} \quad y_p = \text{softmax}(z_3) \text{ and}$$
$$z_3 = V \cdot h_3 + b_y$$



Backpropagation Through Time (BPTT)

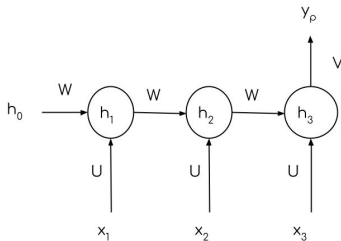


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$$\textcircled{2} \quad y_p = \text{softmax}(z_3) \text{ and } z_3 = V \cdot h_3 + b_y$$

$\textcircled{3}$ Since we know that h_3, b_y are independent of V , we can compute $\frac{\partial L}{\partial V}$ in a single step

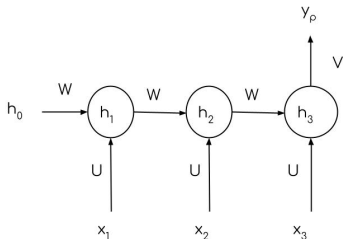


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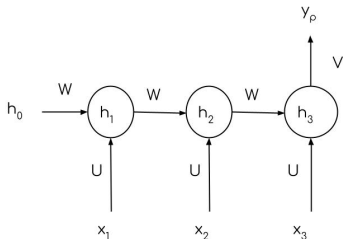
① Let's now consider $\frac{\partial L}{\partial W}$



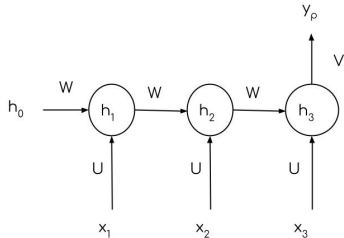
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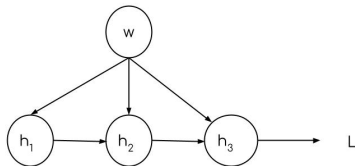
- ① Let's now consider $\frac{\partial L}{\partial W}$
- ② There are multiple ' W 's in the computational graph!



Backpropagation Through Time (BPTT)



- 1 For ease of understanding

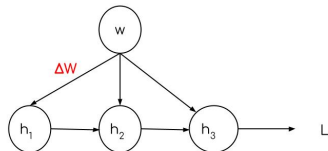


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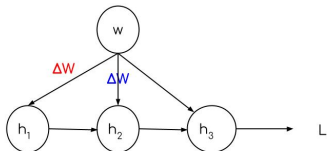
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- ① Δw change in $W \rightarrow$
 $\left(\frac{\partial h_1}{\partial W} \cdot \Delta w \right)$ change in h_1



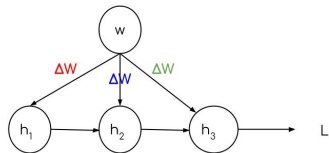
Backpropagation Through Time (BPTT)

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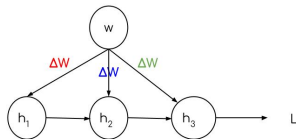
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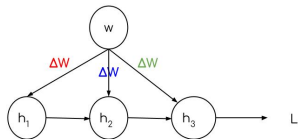
Backpropagation Through Time (BPTT)

$$\textcircled{1} \Delta L = \frac{\partial L}{\partial h_1} \cdot \Delta h_1 + \frac{\partial L}{\partial h_2} \cdot \Delta h_2 + \frac{\partial L}{\partial h_3} \cdot \Delta h_3$$



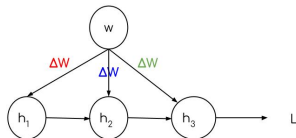
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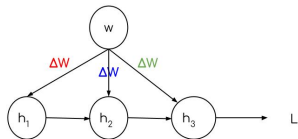
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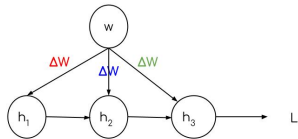
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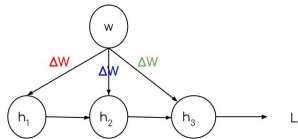
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Backpropagation Through Time (BPTT)

- ① $\Delta L =$
 $\frac{\partial L}{\partial h_1} \cdot \Delta h_1 + \frac{\partial L}{\partial h_2} \cdot \Delta h_2 + \frac{\partial L}{\partial h_3} \cdot \Delta h_3$
- ② $\frac{\partial L}{\partial W} = \frac{\partial L}{\partial h_1} \frac{\partial h_1}{\partial W} + \frac{\partial L}{\partial h_2} \frac{\partial h_2}{\partial W} + \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial W}$
- ③ $\frac{\partial L}{\partial h_3} = \frac{\partial L}{\partial y_p} \frac{\partial y_p}{\partial h_3}$
- ④ $\frac{\partial L}{\partial h_2} = ?$
- ⑤ $\frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2}$
- ⑥ $\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_2} \frac{\partial h_2}{\partial h_1} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1}$



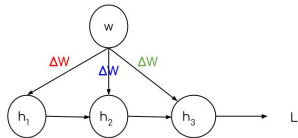
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Backpropagation Through Time (BPTT)

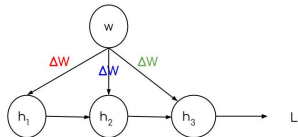
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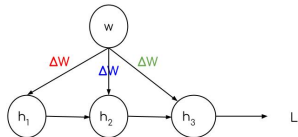
$$③ \quad \frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2}$$

$$④ \quad \frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_1} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1}$$

$$⑤ \quad \frac{\partial L}{\partial W} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1} \frac{\partial h_1}{\partial W} + \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial W} + \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial W}$$

⑥

$$\frac{\partial L}{\partial W} = \sum_{k=1}^3 \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_k} \frac{\partial h_k}{\partial W}$$



Backpropagation Through Time (BPTT)

$$① \quad \frac{\partial L}{\partial W} = \frac{\partial L}{\partial h_1} \frac{\partial h_1}{\partial W} + \frac{\partial L}{\partial h_2} \frac{\partial h_2}{\partial W} + \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial W}$$

$$② \quad \frac{\partial L}{\partial h_3} = \frac{\partial L}{\partial y_p} \frac{\partial y_p}{\partial h_3}$$

$$③ \quad \frac{\partial L}{\partial h_2} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2}$$

$$④ \quad \frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_1} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1}$$

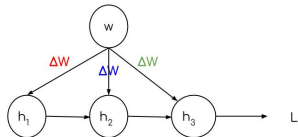
$$⑤ \quad \frac{\partial L}{\partial W} = \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial h_1} \frac{\partial h_1}{\partial W} + \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial h_2} \frac{\partial h_2}{\partial W} + \frac{\partial L}{\partial h_3} \frac{\partial h_3}{\partial W}$$

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⑦

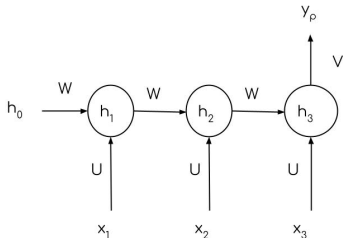
$$\frac{\partial L}{\partial W} = \sum_{k=1}^3 \frac{\partial L}{\partial h_3} \left(\prod_{j=k+1}^3 \frac{\partial h_j}{\partial h_{j-1}} \right) \frac{\partial h_k}{\partial W}$$



Backpropagation Through Time (BPTT)



① Similarly $\frac{\partial L}{\partial U}$

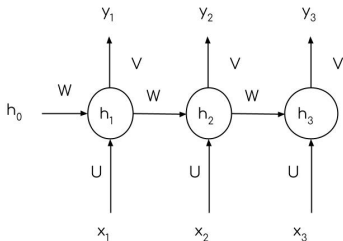


Backpropagation Through Time (BPTT)



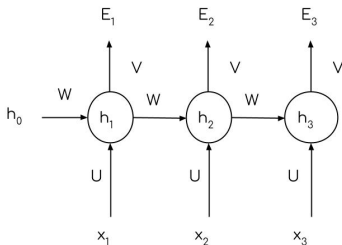
భారతీయ పాఠశాల
భారతీయ ప్రాచీనతా సంస్థ
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- 1 Consider a many-to-many variant RNN (e.g. PoS tagging)



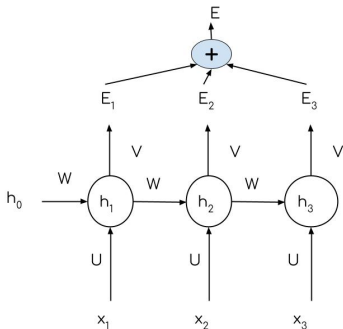
Backpropagation Through Time (BPTT)

- 1 Consider a many-to-many variant RNN (e.g. PoS tagging)
- 2 Full sequence is one training example (although there is an error computed at each time step)



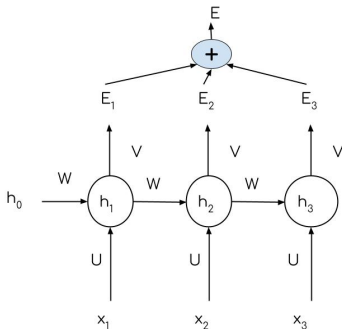
Backpropagation Through Time (BPTT)

- 1 Consider a many-to-many variant RNN (e.g. PoS tagging)
- 2 Total error is the sum of errors at each time step (need not always!)



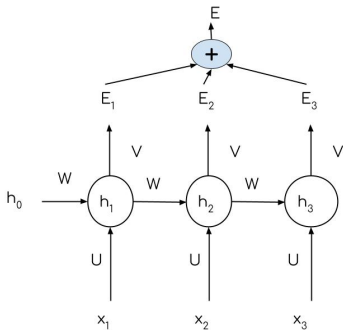
Backpropagation Through Time (BPTT)

- 1 Initialize all derivatives to zero



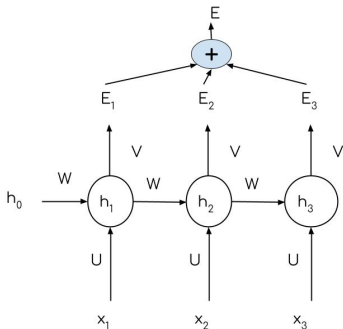
Backpropagation Through Time (BPTT)

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Backpropagation Through Time (BPTT)

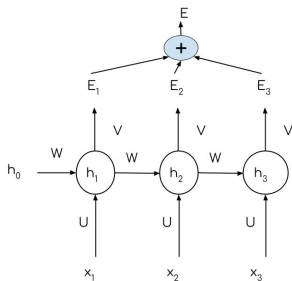
- 1 Initialize all derivatives to zero
- 2 for $t = T$ (in this case 3) till 1
- 3 go on computing the gradients backward



Backpropagation Through Time (BPTT)



భారతీయ టెక్నోలాజికల్ డిజైన్ సెంటర్ హైదరాబాద్
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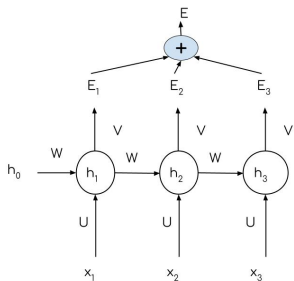


1

Backpropagation Through Time (BPTT)



భారతీయ టెక్నోలాజికల్ డిజైన్ సెంటర్ హైదరాబాద్
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1

Backpropagation Through Time (BPTT)



భారతీయ టెక్నోలాజికల్ విజ్ఞాన సంస్థ హైదరాబాద్
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- ① At times, sequences can be quite lengthy!

Backpropagation Through Time (BPTT)

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Backpropagation Through Time (BPTT)

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- ④ Leads to Vanishing Gradient problem!

Backpropagation Through Time (BPTT)

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- ④ Leads to Vanishing Gradient problem!
- ⑤ No impact of earlier time steps at later times (**difficult to learn long-term dependencies!**)

Backpropagation Through Time (BPTT)



భారతీయ టెక్నోలాజికల్ హైదరాబాద్
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- 1 In some cases $\left(\prod_{j=k+1}^3 \frac{\partial h_j}{\partial h_{j-1}} \right)$ may lead to exploding gradients

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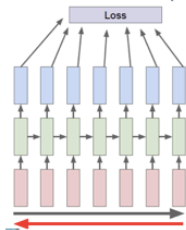
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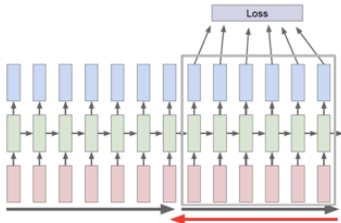
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- ② But, not much of an issue
 - Easy to diagnose (NaN)
 - Gradient clipping
- ③ Better initialization, Regularization, short time sequences (Truncation)

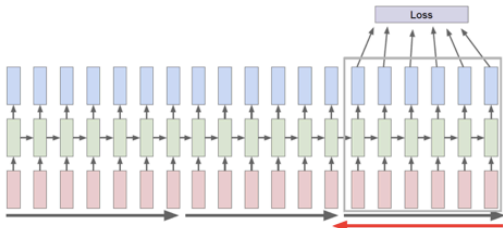
Backpropagation Through Time (BPTT)



(a)



(b)



(c)

Truncated BPTT (CS231n)

① Bidirectional RNNs

Next

- ① Bidirectional RNNs
- ② Stability analysis

Next

- ① Bidirectional RNNs
- ② Stability analysis
- ③ LSTMs